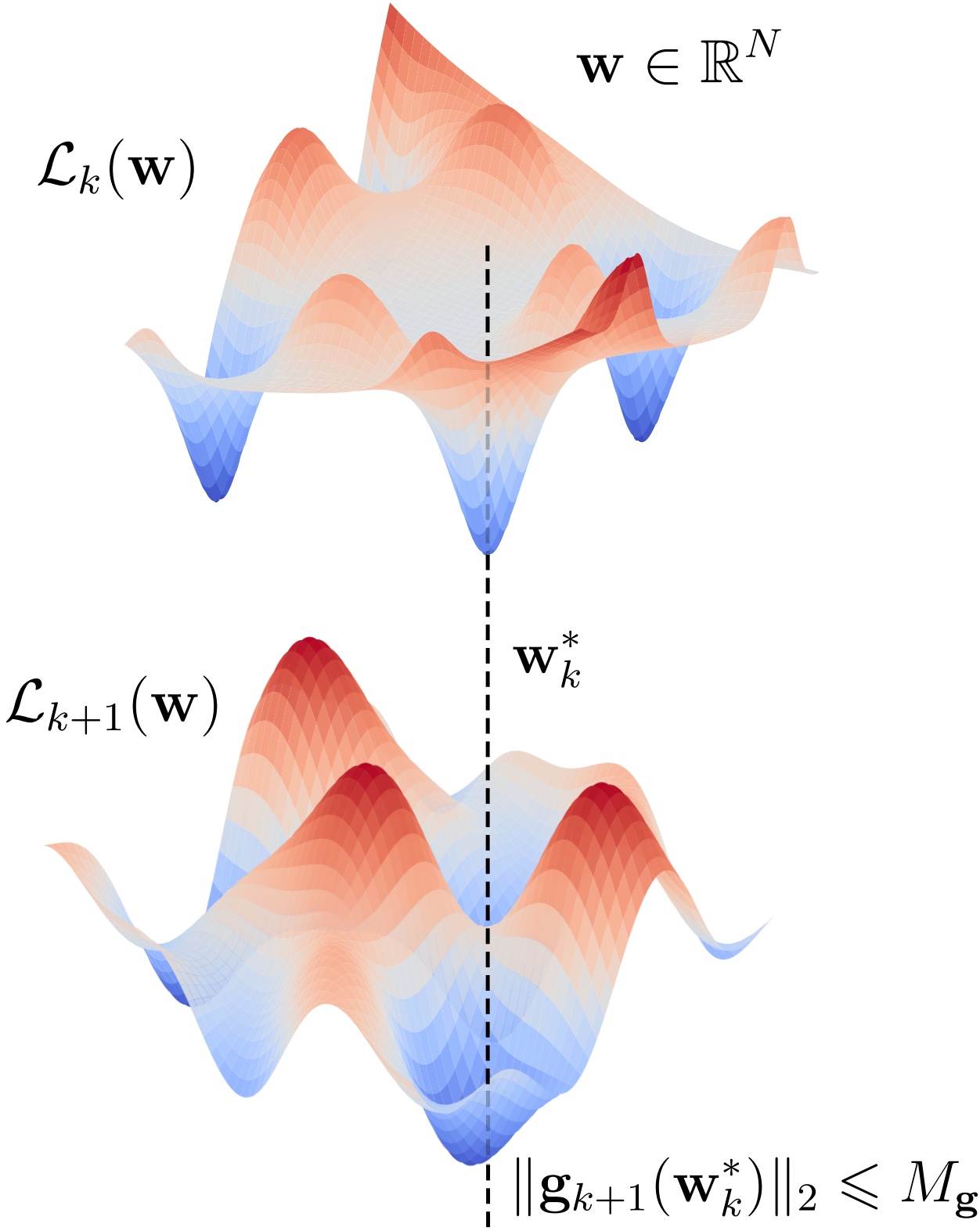
$\mathbf{w} \in \mathbb{R}^N$
$\mathcal{L}_k(\mathbf{w})$
$\mathbf{w}_k^*$
$\mathcal{L}_{k+1}(\mathbf{w})$
$\|\mathbf{g}_{k+1}(\mathbf{w}_k^*)\|_2 \leqslant M_{\mathbf{g}}$